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Review

A comprehensive review of hydrogen integrated hybrid renewable energy systems: Configurations, models, simulation and optimization with artificial intelligence

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ABSTRACT

This work presents a comprehensive review of hydrogen-based hybrid renewable energy systems (HRESs), covering mathematical models, simulation and artificial intelligence (AI)-driven optimization approaches. Emphasizing the potential of hydrogen as an energy carrier to deepen renewable energy integration, especially in solar and wind HRESs, this review systematically details mathematical models for various renewable generation and storage systems, serving as a structured reference for researchers. Given the complexity of HRES modeling, this work provides insights into different modeling software and optimization algorithms, with a particular focus on artificial intelligence methods. The integration of software and artificial intelligence promises to solve complex modeling and optimization challenges with potential applications in different environments. Future directions suggest that the physical model-assisted AI framework, which embeds physical principles within AI models, holds promise for enhancing prediction accuracy and reliability in HRES applications. This framework, especially when combined with stochastic optimization, offers a potential pathway to address challenges in data availability and computational complexity, supporting the effective design and optimization of hydrogen-based HRESs for real-world applications. The overall findings will help improve the design and optimization of hydrogen-based hybrid renewable energy systems for practical implementation.

1. Introduction

Alternative clean, renewable energy sources offer solutions to the energy crisis. Following the EU's goal of carbon neutrality by 2050, secure, sustainable and competitive energy systems must be established to meet Net Zero Emissions targets (Potrč et al., 2021). Similarly, China's ambition for carbon neutrality by 2060 requires economic shifts toward zero-emission electricity, renewable energy expansion, and carbon capture and storage (Zhang and Hanaoka, 2021). Hydrogen has the potential to decarbonize and integrate electricity and natural gas systems, fostering the development of coupled electricity-hydrogen-gas networks (Jiang et al., 2024; Johnson et al., 2025). With the rapid growth of renewable energy, water electrolysis is a widely adopted method for producing green hydrogen. Additionally, hydrogen serves as a long-term energy buffer, balancing power supply and demand (Risco-Bravo et al., 2024). Research on renewable energy and hydrogen is thus essential and urgent.

Renewable energy sources are essential for protecting the environment and promoting sustainable economic and social development. Numerous studies have concentrated on a particular type of renewable energy, such as photovoltaic (PV), wind and hydropower (AbdelHady, 2019; Guo et al., 2018). However, a single-source renewable energy system (SRES) frequently has an excessive scale to deliver enough power to the load because renewable energy sources are random and

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Abbreviation

ACO	Ant Colony Optimization
AI	Artificial Intelligence
ANN	Artificial Neural Network
BBO	Biogeography-Based Optimization
COE	Cost of Energy
CRF	Capital Recovery Factor
DL	Deep Learning
DT	Decision Trees
FA	Firefly Algorithm
FC	Fuel Cell
GA	Genetic Algorithm
GHE	Greenhouse Gas Emission
HDI	Human Development Index
HRES	Hybrid Renewable Energy Systems
HS	Harmony Search
LCC	Life Cycle Cost
LOEE	Loss of energy expected
LPS	Loss of Power Supply
LPSP	Loss of Power Supply Probability
LSTM	Long Short-Term Memory
MILP	Mixed-Integer Linear Programming
ML	Machine Learning
NPC	Net Present Cost
NSGA	Non-dominated Sorting Genetic Algorithm
PSO	Particle Swarm Optimization
PV	Photovoltaic
RF	Random Forest
SA	Simulated Annealing
SVM	Support Vector Machine
WT	Wind Turbines
Nomenclature	
a_1, a_2	idealized coefficients of the two diodes
a_{H_2O}	water activity between anode and electrolyte
A	collector aperture area, m^2, area swept by the turbine blade, m^2
A_{PV}	area of PV, m^2
C_p	power coefficient of the turbine
E_0	standard potential, V
$E_{L(t)}$	load demand in hour t, kWh
E_{Nernst}	open-circuit voltage, V
$f_{max_{SE}}$	element of maximum surplus energy
$f_{max_{P_{LOAD}}}$	element to increase yearly AC load
P_{SE}	yearly surplus energy, kWh/yr
G	incident solar radiation perpendicular to the surface, W/m^2; radiation intensity on the surface of the solar cell, W/m^2
GHE_i	greenhouse gas emission of the system component i
G_{stc}	radiation intensity under standard test conditions, W/m^2
i	current density, A/m^2
i_0	exchange current density, A/m^2
i_L	limiting current density, A/m^2
ir	annual interest rate
I	current in the circuit, A; actual interest; output current, A
I_{01}, I_{02}	reverse saturation currents of the two diodes, A
I_b	direct radiation, W/m^2
I_d	scattered radiation, W/m^2
I_{pv}	PV current, A
I_{sc}	short-circuit current, A
K_1	temperature coefficient of the short-circuit current
K_v	temperature coefficient of open-circuit voltage
LOE_i	amount of loss of energy, kWh
n_{gas}	hydrogen flow rate, mol/s
n_{poly}	polytropic index
n_{st}	number of moles of hydrogen inside the storage tank, mol
N	lifetime number
OMC_i	annual operation and maintenance cost of component i
$p_{c,in}$	pressure at the inlet of the compressor, Pa
$p_{c,out}$	pressure at the outlet of the compressor, Pa
p_{st}	pressure inside the storage tank, Pa
P_{comp}	power required by the compressor, W
P_e	electrical power produced by the wind turbine, W; excess power, kW
P_{er}	rated electrical power output of wind turbine, W
P_{g_Elec}	amount of electricity sent to the electrolyzer by the PV/WT power generation system, kWh
P_m	mechanical power output of the turbine, W
P_{sj}	hourly power output on an average day of jth month, W
P_t	power of transmission system of the wind turbine, W
P_{H_2}	partial pressures of hydrogen
P_L	hourly power consumption, kWh

(*continued*)

$P_{LY/C}$	energy yearly electricity consumption per capita, kWh/yr/person
P_{O_2}	partial pressures of oxygen
P_{SE}	yearly surplus energy of the system, kWh/yr
P_W	wind power, W
Q_R	radiation loss, W
R	universal gas constant, 8.314 J/(mol•K); total electric resistance, Ω
RC_i	sum of the replacement costs of component i
R_d	correction factor orrection factor for diffuse radiation
R_{WT}	radius of the turbine blade, m
R_ρ	ground reflection
t_m	proton exchange membrane thickness, m
T	operating temperature, K; ambient temperature, °C
T_c	monthly average temperature of the battery, °C
T_{el}	cell temperature, K
T_r	theoretical temperature of the battery, °C
T_{stc}	ambient temperature under standard test conditions, °C
U_L	heat loss coefficient per unit area of the photovoltaic array
U_{T1}, U_{T2}	terminal voltages of the two diodes, V
v	wind velocity, m/s
v_c	cut-in speed of the wind turbine, m/s
v_f	cut-out (or furling) speed of the wind turbine, m/s
v_r	rated speed of the wind turbine, m/s
V_{act}	activation overvoltage, V
V_{con}	concentration overvoltage, V
V_{ohm}	ohmic overvoltage, V
V_{st}	volumetric capacity of the storage tank, m^3
Greek Symbols	
α	solar absorption rate of the photovoltaic array; charge transfer coefficient
ΔG_f	Gibbs free energy
η_{PVT}	total efficiency of the photovoltaic thermal hybrid solar collector
η_{comp}	compressor efficiency
η_{elec}	efficiency of the electrolyzer
η_{fc}	electric output energy divided by the hydrogen input energy of the system
η_g	generator efficiency
η_m	transmission efficiency
η_{pc}	efficiency of the power regulator
η_r	theoretical photoelectric conversion rate
η	conversion efficiency for the hybrid PV/TE system
η_{HUMAN}	number of humans who consume the generated power
ρ	electric resistivity, Ω·m; ground reflectivity; air density, kg/m^3
σ_m	conductivity of the membrane, S/m
τ	solar penetration rate of the photovoltaic array

intermittent (Prebeg et al., 2016). Therefore, hybrid renewable energy systems (HRESs) have drawn increased attention. By integrating several renewable energy sources with other energy storage and power-generating technologies, HRESs are designed to work in concert with each renewable energy source to meet the load requirement, especially in remote areas (Ram et al., 2021). Batteries and hydrogen technologies have shown effective short- and long-term storage methods. In the planning and design process of HRES, multiple objectives usually need to be considered, such as cost minimization and high reliability of power supply. An optimal sizing combination is crucial for achieving higher reliability at the lowest costs. Therefore, modeling, simulation and optimization of HRES are indispensable, especially regarding the issues of technology, economy and environment. These are key steps to verify the system's feasibility (Sharma et al., 2019). The biggest problem in designing HRES is selecting the optimal configuration to ensure the power supply system has the highest reliability, the lowest power generation cost, and minimal environmental pollution. Anoune et al. (2018) introduced the most commonly used optimization sizing techniques of HRES in remote areas to achieve the best trade-off consistency between power reliability and hybrid power system costs.

In addition to summarizing the technical, economic and environmental indicators of HRES, Cuesta et al.Cuesta et al. (2020) also emphasized that social factors should be considered when designing renewable energy systems. They advocated that relevant developers should increase social parameters, such as the human development index (HDI) and job creation, when using software tools. Siddaiah and Saini (2016) summarized the studies on the planning, configuration, modeling and optimization technology of HRES for off-grid applications. They divided the optimization methods for renewable energy into three categories: classical, artificial intelligence, and hybrid techniques. Eriksson and Gray (2017) summarized the related works on optimizing and integrating hybrid renewable energy and hydrogen fuel cell systems. They proposed a linear combination of multi-objective optimization indicators for technical, economic, environmental, and sociopolitical factors into a single method of goal optimization. Considering the development prospects of hydrogen energy, Laimon and Yusaf (2024) explore the latest progress and challenges of integrated renewable energy-driven hydrogen systems.

Although the studies were conducted on optimal planning, their research contents are relatively separate and isolated. They consider either the control strategy (Hossain et al., 2023), optimization of the off-grid system (Siddaiah and Saini, 2016) or the on-grid system (He et al., 2023; Kurukuru et al., 2021), the optimization of PWHS (Anoune et al., 2018), or the solar PV system (Kurukuru et al., 2021). Artificial intelligence (AI), especially deep learning algorithms, can effectively implement the design, forecasting, control, optimization, and maintenance of power systems (Bansal, 2022; George Davies et al., 2023). However, there are relatively few studies on AI technology.

Given the potential of AI technology, this study provides a systematic review of recent developments in optimization, drawing on the latest research perspectives. The study aims to address issues related to hydrogen-integrated HRES by conducting a comprehensive review of system configurations, mathematical modeling, energy storage methods, simulation, and optimization using artificial intelligence. The review follows a systematic framework for literature selection and classification. Specifically, peer-reviewed journal articles indexed in Web of Science (SCI), Scopus, and IEEE Xplore were systematically searched using the keywords "hybrid renewable energy system (HRES)", "optimization", "hydrogen integration", "energy management", "machine learning" and "artificial intelligence". Publications from 2010 to 2025 were considered to include both foundational and state-of-the-art research. Studies were included if they (a) focused on modeling or optimization of hydrogen-based HRES, (b) presented quantitative results, and (c) were published in reputable international journals. Conference abstracts, short reviews, and studies lacking quantitative validation were excluded.

2. Models of hydrogen integrated hybrid renewable energy systems

Hydrogen-integrated hybrid renewable energy systems (HRESs) have emerged as an effective solution to the intermittency and variability of renewable energy sources. By combining hydrogen production, storage, and utilization within a hybrid multi-energy architecture, these systems can deliver continuous, reliable, and low-carbon electricity and heat (Ge et al., 2025). Hydrogen not only serves as a short-term buffer during low wind or solar output but also enables seasonal energy storage, allowing excess energy generated in high-production periods (e.g., summer for PV or windy seasons) to be stored and used later during periods of scarcity. Furthermore, hydrogen facilitates multi-sector coupling, linking the power system with transportation, industrial processes, and heating, thereby positioning itself as a key pillar in future integrated energy systems.

Fig. 1 presents a schematic of a hydrogen-based energy system, in which electricity from renewable energy sources and conventional power plants is supplied to the grid and partially used for water electrolysis. The generated hydrogen can be stored and later converted back to electricity via fuel cells or gas turbines (Power-to-Power), supplied to hydrogen refueling stations for mobility applications (Power-to-Mobility), combined with CO_2 to produce synthetic methane via methanation (Power-to-Gas), or used directly as an industrial feedstock (Power-to-Feedstock). This versatility underscores hydrogen's essential role not only in energy storage but also in reducing carbon emissions across multiple sectors.

Based on this system framework, this section presents the fundamental physical and mathematical models of the major subsystems within hydrogen-integrated HRESs, including photovoltaic (PV) generation, wind energy conversion, electrolysis, hydrogen storage, and fuel cell technologies. The objective is to establish the core mathematical formulations that describe the physical principles and operating mechanisms of these components. It should be emphasized that the focus herein is not on model comparison or performance assessment. Instead, the models included are widely adopted in the literature and provide the theoretical foundation for subsequent system simulation, techno-economic evaluation, and multi-objective optimization. The implementation of these formulations in simulation platforms and their integration with optimization algorithms are discussed in detail in Section 3.

2.1. Photovoltaic (PV) power generation model

The input of the PV system is solar radiation, which includes direct, diffuse and reflected radiation. The total radiation I_T can be expressed as (Deshmukh and Deshmukh, 2008):

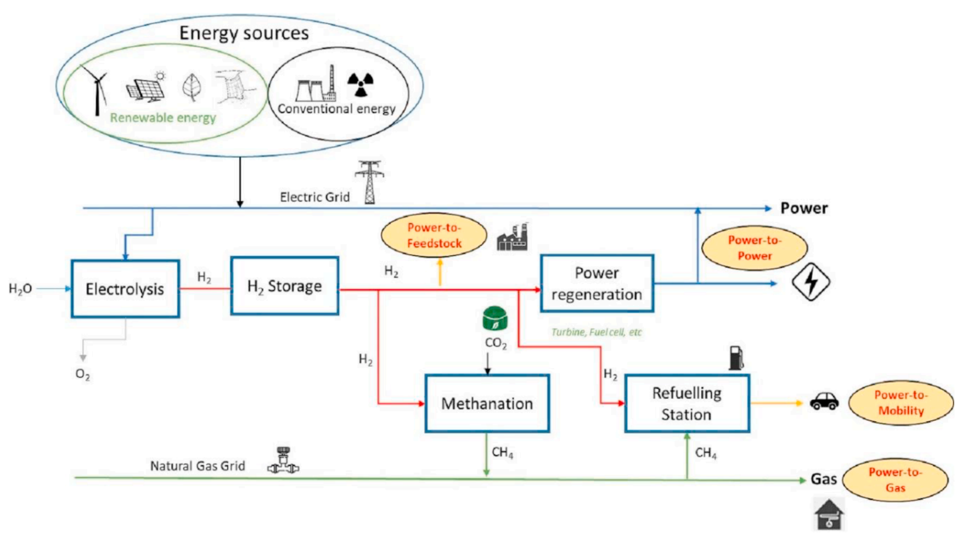

Fig. 1. Schematic of the hydrogen value chain including Power to X applications (Luo et al., 2023).

$$I_T = I_b R_b + I_d R_d + (I_b + I_d) R_r \tag{1}$$

$$R_r = \rho \frac{1 - \cos \beta}{2} \tag{2}$$

$$R_d = \rho \frac{1 + \cos \beta}{2} \tag{3}$$

where T is the inclined plane, I_b and I_d are the direct radiation and scattered radiation, W/m^2; R_b, R_d, R_ρ are correction factors for direct radiation, diffuse radiation and ground reflection; ρ is the ground reflectivity; β is the tilt angle (Kolhe et al., 2003).

The output power P_{sj} of the photovoltaic system with the total radiation amount of I_T in month j can be expressed as:

$$P_{sj} = I_{Tj} \eta A_{\mathrm{PV}} \tag{4}$$

$$\eta = \eta_m \eta_{\mathrm{pc}} P_f \tag{5}$$

$$\eta_m = \eta_r [1 - \beta(T_c - T_r)] \tag{6}$$

where A_{PV} is the area, m^2; η_r is the theoretical photoelectric conversion rate; η_{pc} is the efficiency of the power regulator; P_f is the fill factor; β is the photovoltaic array efficiency temperature coefficient; T_r is the theoretical temperature of the battery, °C; T_c is the monthly average temperature of the battery, °C.

According to the energy conservation equation:

$$\alpha \tau I_T = \eta_m \tau I_T + U_L (T_c - T_a) \tag{7}$$

$$T_c = T_a + \frac{\alpha \tau}{U_L} I_T \tag{8}$$

$$U_L / \alpha \tau = I_{T,\mathrm{NOCT}} / \left(\mathrm{NOCT} - T_{a,\mathrm{NOCT}}\right) \tag{9}$$

where U_L is the heat loss coefficient per unit area of the photovoltaic array; α is the solar absorption rate of the photovoltaic array; τ is the solar penetration rate of the photovoltaic array; T_a is the ambient temperature, °C. NOCT is the nominal operating cell temperature.

A dual diode model generally expresses the solar cell model (Ishaque et al., 2011). The simplified circuit is shown in Fig. 2.

Where I_{01} and I_{02} are the reverse saturation currents of the two diodes, A; U_{T1} and U_{T2} are the terminal voltages of the two diodes, V, and a_1 and a_2 are the idealized coefficients of the two diodes. The diode currents are calculated using the Shockley equation.

The output current I is expressed as:

$$I = I_{\mathrm{PV}} - I_{01} \left[\exp\left(\frac{U + IR_S}{a_1 U_{T1}} \right) - 1 \right] - I_{02} \left[\exp\left(\frac{U + IR_S}{a_2 U_{T2}} \right) - 1 \right] - \frac{U + IR_S}{R_p} \tag{10}$$

The PV current I_{PV} is expressed as:

$$I_{\mathrm{PV}} = (I_{\mathrm{sc}} + K_1 \Delta T) \frac{G}{G_{\mathrm{stc}}} \tag{11}$$

$$\Delta T = T - T_{\mathrm{stc}} \tag{12}$$

where I_{sc} is the short-circuit current, A; K_1 is the temperature coefficient; T is the ambient temperature, °C; G is the radiation intensity on the surface of the solar cell, W/m^2; T_{stc} (25°C) and G_{stc} (1000 W/ m^2) are the ambient temperatures and radiation intensity under standard test conditions.

Reverse saturation currents I_{01}, I_{02} are expressed as:

$$I_{01} = I_{02} = \frac{(I_{\mathrm{sc}} + K_1 \Delta T)}{\exp[(U_{\mathrm{oc}} + K_V \Delta T)/((a_1 + a_2)/p) U_T] - 1} \tag{13}$$

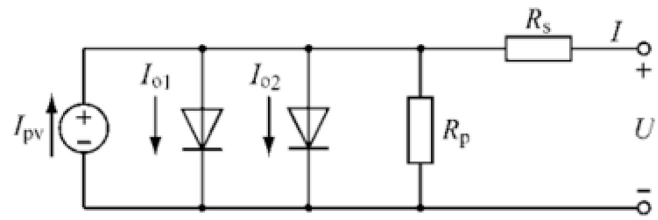

Fig. 2. Double diode equivalent circuit model of a photovoltaic cell in a photovoltaic array.

where K_{V} is the temperature coefficient of open-circuit voltage, which can be obtained in the data sheet provided by the manufacturer and U_{oc} is the open-circuit voltage, V.

2.2. Wind power generation model

Wind turbines transform kinetic energy into mechanical energy and further to electrical energy based on fluid dynamics and electromagnetic induction principles. When wind turbine (WT) blades generate torque under the action of airflow to drive the rotation of the wind turbine, the co-axially connected generators rotate together, thus realizing the conversion of wind to electricity (Hall et al., 2011).

According to the aerodynamics principle, wind power (P_W) can be expressed as:

$$P_W = \frac{1}{2} \rho A v^3 \tag{14}$$

where ρ is the air density, kg/m^3; A is the area swept by turbine blades,m^2; v is the wind velocity, m/s.

Then, the mechanical power output of the turbine (P_m) is:

$$P_m = C_p P_W \tag{15}$$

where C_p is the turbine power coefficient.

Next, the power output of the transmission system of the wind turbine (P_t) and the electrical power (P_e) are respectively:

$$P_t = \eta_m P_m \tag{16}$$

$$P_e = \eta_g P_t \tag{17}$$

where η_m and η_g are the transmission efficiency and generator efficiency respectively.

Based on fundamental equations of power, P_e can be expressed as follows:

$$P_e = \begin{cases} 0 \quad 0 \le v < vc \\ \frac{1}{2} \eta_m \eta_g C_p \rho A v^3 \ v_c \le v < v_r \\ P_{e,r} \ v_r \le v \le v_f \\ 0 \ v > v_f \end{cases} \tag{18}$$

where v_c is the cut-in speed, m/s; v_r is the rated speed, m/s; v_f is the cut-out (or furling) speed, m/s; $P_{e,r}$ is the rated electrical power output of a wind turbine, W.

Based on a presumed shape of the power curve, P_e can be expressed as follows:

$$P_e = \begin{cases} 0 \quad 0 \le v < vc \\ P_{e,r} \frac{v^n - v_c^n}{v_r^n - v_c^n}, n = 1,2,3 \quad v_c \le v < v_r \\ P_{e,r} \ v_r \le v \le v_f \\ 0 \ v > v_f \end{cases} \tag{19}$$

2.3. Models of hydrogen energy storage

As shown in Table 1, Hydrogen can be stored using metal hydrides, ammonia, liquid organic hydrogen carriers (LOHCs), or compressed gas, each with distinct advantages and limitations. Metal hydrides and advanced solid-state materials (e.g., MOFs, carbon-based nanomaterials, borohydrides) provide compact, safe, and high-density storage, yet their high cost, weight, and slow kinetics hinder large-scale deployment. Ammonia offers high energy efficiency and the potential for direct utilization, but its synthesis, decomposition, and toxicity present significant challenges. LOHCs, including formic acid and alcohols, are compatible with existing fuel infrastructures and allow safer handling, though hydrogen release is energy-intensive and requires highly active catalysts. Among these methods, compressed hydrogen gas has emerged as the most widely studied in hydrogen-integrated renewable energy systems, as highlighted by Irham et al. (2024), due to its simplicity, efficient volumetric density, and practicality for transportation and storage. The disadvantages of compressed hydrogen storage are compressibility, safety issues, low energy density and transportation costs. However, despite these disadvantages, compressed hydrogen storage is becoming a commercial reality for fuel cell vehicles and refueling stations. Compressed hydrogen storage is technically feasible when powered by fuel cells. The selection of storage technology critically impacts system efficiency, safety, and cost-effectiveness, and overcoming the trade-offs among energy density, operational feasibility, and economic considerations remains essential for the commercialization of hydrogen-based renewable energy systems (Ajanovic et al., 2024).

2.3.1. Compressed hydrogen storage

Compressed hydrogen storage is the most technically mature and widely used method, where hydrogen gas is stored in high-pressure vessels, typically at pressures up to 35–70 MPa. Its advantages include rapid charging and discharging, as well as relatively simple system integration; however, it requires robust tanks and consumes significant energy during compression. In modeling compressed hydrogen storage, the work required to compress hydrogen to a target pressure is commonly represented as a polytropic process, which characterizes the relationship between pressure, volume, and temperature during the compression process. To determine the actual compression work, compressor efficiency is introduced to account for real-world energy losses (Deshmukh and Boehm, 2008):

$$P_{comp} = \frac{n_{\mathrm{poly}}}{n_{\mathrm{poly}} - 1} n_{\mathrm{gas}} RT \left[1 - \left(\frac{p_{c,out}}{p_{c,\mathrm{in}}} \right)^{\left(n_{\mathrm{poly}}-1\right)/n_{\mathrm{poly}}} \right] \frac{1}{\eta_{\mathrm{comp}}} \tag{20}$$

where P_{comp} is the power required by the compressor, W; η_{comp} is the compressor efficiency, n_{poly} is the polytropic index, n_{gas} is the molar flow rate of hydrogen, mol/s; T is the operating temperature, K; $p_{c,\mathrm{in}}$ and $p_{c,\mathrm{out}}$ are the pressures at the compressor inlet and outlet, Pa.

The hydrogen mass of the storage tank can be determined using the following expression (Luo et al., 2024):

$$m = \frac{PV_{\mathrm{st}} M_{\mathrm{H2}}}{ZRT} \tag{21}$$

where P is the pressure inside the storage tank, Pa; Z is the compressibility factor, M_{H2} is the molar mass of hydrogen; V_{st} is tank volume, m^3. In our previous studies, depending on how the geometric structure of the storage tank is treated, three types of lumped parameter models have been developed: the single-zone model (hydrogen only), the dual-zone model (hydrogen and tank wall), and the triple-zone model (hydrogen, tank wall liner and shell). Detailed formulations of the thermodynamic and heat transfer models can be found in our earlier work (Luo et al., 2024; Xiao et al., 2024).

2.3.2. Physical adsorption hydrogen storage

Physisorption-based hydrogen storage primarily relies on van der Waals interactions between hydrogen molecules and the micropore walls of the adsorbent. The storage capacity is strongly dependent on the microporous structure and surface energy distribution of the material. Typical adsorbents include carbon-based materials, metal-organic frameworks (MOFs), covalent organic frameworks (COFs), graphene, and zeolites (Wen et al., 2025).

The classical Dubinin-Astakhov (D-A) model was originally developed to describe the micropore filling of subcritical gases in porous adsorbents. Its general form is expressed as (Xi et al., 2025):

$$\theta = \frac{n_a}{n_{max}} = \exp\left[- \left[\frac{A}{\varepsilon} \right]^m \right] \tag{22}$$

$$A = RT \ln\left(\frac{P_0}{P} \right) \tag{23}$$

where θ denotes the degree of micropore filling, n_a is the absolute adsorption amount, n_{max} is the maximum adsorption capacity, A is the adsorption potential related to free energy variation, ε is the characteristic adsorption energy, m is an empirical parameter reflecting surface heterogeneity, P_0 is the saturation vapor pressure at adsorption temperature T, and P is the equilibrium pressure.

While the D-A model provides a good description for subcritical adsorption, its applicability becomes limited under hydrogen storage conditions, where the operating temperature and pressure typically

Table 1
Advantages and disadvantages of different hydrogen storage methods (Khaja Muhieitheen and Shahul Hameed, 2025).

Storage method	Description	Advantages	Disadvantages	Efficiency (%)	Estimated Cost (USD/kg H_2)
Compressed Gas	• Gaseous hydrogen is stored or transported with pressures between 200 and 500 bar	•Well-established technology, especially on a relatively small scale •Pressurized hydrogen storage method can quickly fill and discharge	•Delivers hydrogen at high pressure •Store at low temperature or high pressure	40%–55%	2.0–5.0
Liquid Hydrogen	•Liquid hydrogen is stored at − 253°C or normal temperature under pressure	•The low-pressure liquid hydrogen storage system has a low cost •High liquid density and storage efficiency	•Cooling systems are needed	25%–45%	3.5–8.0
Material	• Divided into adsorption and absorption methods • Hydrides and alloys	• Large amounts of hydrogen can be stored in a small volume	• Storage materials have weight and volume	50%–70%	10.0–25.0
Liquid Ammonia Storage	• Hydrogen can be stored as ammonia (NH_3) and released by decomposition when needed	• High hydrogen content • Efficient release of hydrogen • Suitable for large-scale storage and transport	• Energy-intensive to extract H_2 • Toxicity of ammonia • Complex transportation requirements	40%–65%	1.5–4.5

exceed the critical point of hydrogen. In this regime, the physical meaning of n_{max} and P_0 is diminished, and the assumption of temperature invariance of ε is no longer valid, since ε tends to vary approximately linearly with temperature. To overcome these drawbacks, a modified D-A model has been developed. In this revised form, n_{max} and P_0 are treated as constant fitting parameters to preserve simplicity, whereas enthalpy and entropy factors are introduced to capture the temperature dependence of the characteristic adsorption energy. The modified expression is given as (Richard et al., 2009):

$$n_a = n_{max}\exp\left[-\left(\frac{RT}{\alpha+\beta T}\right)^m \ln^m\left(\frac{P_0}{P}\right)\right] \tag{24}$$

where R is the universal gas constant, and α and β are the enthalpy and entropy factors, respectively. This modification extends the applicability of the original D-A equation and provides a more accurate description of hydrogen physisorption, particularly under cryogenic and high-pressure conditions. In our previous work, the modified D-A model has been successfully applied to describe the adsorption of H_2 on activated carbon (Tong et al., 2016; Xiao et al., 2012).

2.3.3. Metal hydride hydrogen storage

Metal hydride hydrogen storage relies on the reversible absorption and desorption of hydrogen molecules within the crystal lattice of metals or alloys. When hydrogen gas interacts with a metallic material, hydrogen molecules dissociate into atoms and diffuse into interstitial sites of the lattice, forming metal hydrides. The process is reversible under appropriate temperature and pressure conditions, enabling a high volumetric hydrogen density. The thermodynamic behavior of metal hydrides is generally described by the Van't Hoff equation (Alleori et al., 2025):

$$\ln\left(\frac{P_{eq}}{P_{ref}}\right) = \frac{\Delta H}{RT} - \frac{\Delta S}{R} \tag{25}$$

where P_{eq} is the reaction equilibrium pressure, P_{ref} represents the reference pressure, ΔH is the enthalpy change of the hydriding reaction, ΔS is the entropy change, and T is the absolute temperature. This relationship highlights that the stability of metal hydrides is primarily governed by the reaction enthalpy and entropy.

The hydrogen storage capacity is usually expressed as the weight fraction (wt.%):

$$\text{wt.\%} = \frac{M_{H2}}{M_{H2}+M_{MH}+M_{shell}} \times 100 \tag{26}$$

where wt.% is the storage capacity of the system including tank shell, M_{H2} is the mass of absorbed hydrogen, M_{MH} is the mass of the metal or alloy host, M_{shell} is the mass of the tank shell. Representative materials include $LaNi_5$, MgH_2 and complex hydrides such as $NaAlH_4$, each offering specific advantages in terms of reversibility, storage density, and kinetics. Since the hydriding process is accompanied by substantial heat release, efficient thermal management is crucial. In our previous studies, we demonstrated that integrating helical coil heat exchangers or phase change materials (PCMs) into metal hydride storage systems effectively reduced the in-tank temperature rise, thereby improving the overall hydrogen storage capacity (Tong et al., 2019, 2023).

2.4. Electrolyser and fuel cells

An electrolyzer is an electrochemical device that splits water into H_2 and O_2 through electrolysis. It comprises multiple cells connected in series, with each cell containing two electrodes separated by either an aqueous or solid polymer electrolyte. The empirical U-I relationship for an electrolyzer cell is given by (Tamalouzt et al., 2016):

$$U_{cell} = U_{rev} + (r_1 + r_2 T) I_{Elec} / A + k \tag{27}$$

$$k = k_{Elec}\ln\left[\left(k_{T_1} + k_{T_2}/T + k_{T_3}/T^2\right) I_{Elec}/A + 1\right] \tag{28}$$

where U_{cell} is the cell terminal voltage, V; U_{rev} is the reversible portion, V; I_{Elec} is the elecreolyzer current, A; A is the area of the cell electrode, m^2; r_1 and r_2 are the parameters related to Ohmic resistances; k and k_{Elec} are the overvoltage and its coefficient, V; k_{T_1}, k_{T_2} and k_{T_3} are temperature coefficients; T is the electrolyser cell temperature, °C. Electrolyser operating at higher temperatures requires lower terminal voltage under a given current.

For the entire system, the power generation rate at time t is represented by $E_g(t)$ and the load is represented by $E_L(t)$.

If $E_g(t) \geq E_L(t)/\eta_{inv}$, the electrolyser charges the hydrogen tanks and the hydrogen storage quantity can be given by (Maleki and Askarzadeh, 2014; Zhang et al., 2018):

$$SOC_{HT}(t) = SOC_{HT}(t-1) + \left[E_g(t) - \frac{E_L(t)}{\eta_{inv}}\right]\eta_{Elec} \tag{29}$$

If $E_g(t) \leq E_L(t)/\eta_{inv}$, The FC supplies the electrical load and the hydrogen storage quantity at hour t is given by (Maleki and Askarzadeh, 2014; Zhang et al., 2018)

$$SOC_{HT}(t) = SOC_{HT}(t-1) - \left[\frac{E_L(t)}{\eta_{inv}} - E_g(t)\right] \Big/ \eta_{ELec} \tag{30}$$

where η_{inv} is the inverter efficiency and η_{ELect} is the electrolyser efficiency.

The formula for the hydrogen produced by the electrolytic cell is as follows (Fathy, 2016):

$$m_{H_2} = \eta_{Elec} P_{Elec} / 39.4 \tag{31}$$

where P_{Elec} is the amount of electricity sent to the electrolyser by the PV/WT power generation system, kWh.

The electrolyzer converts electrical energy into chemical energy by decomposing water into hydrogen and oxygen. In contrast, the fuel cell electrochemically generates electricity and water from hydrogen and oxygen. Fuel cells are considered an effective technology for combined heat and power production and for alleviating the intermittency of renewable energy systems (Sadeghian et al., 2025). In practice, the output voltage of a fuel cell is lower than its theoretical value due to irreversible polarization losses, also known as overpotential or overvoltage. These losses are mainly classified into activation, ohmic and concentration polarizations:

$$V_{cell} = E_{Nernst} - V_{act} - V_{ohm} - V_{con} \tag{32}$$

where V_{act} is the activation overvoltage, V; V_{ohm} is the ohmic overvoltage, V; V_{con} is the concentration overvoltage, V. Open circuit voltage (E_{Nernst}) is defined as the Nernst Equation:

$$E_{Nernst} = E_0 - \frac{RT_{el}}{2F}\left[\ln\left(\frac{P_{H_2} P_{O_2}{}^{1/2}}{a_{H_2O}}\right)\right] \tag{33}$$

where R is the universal gas constant; P_{H_2} and P_{O_2} are the partial pressures of H_2 and O_2, Pa; T_{el} is the cell temperature, K, and a_{H_2O} is water activity between anode and electrolyte; E_0 is the standard potential, its expression is:

$$E_0 = \frac{\Delta G_f}{2F} \tag{34}$$

where ΔG_f is Gibbs' free energy of formation. The V_{act} is obtained by (Gorgun, 2006):

$$V_{\text{act}} = E_0 + \frac{RT_{\text{el}}}{2\alpha F}\left[\ln\left(\frac{i}{i_0}\right)\right] \tag{35}$$

where α is the charge transfer coefficient, i and i_0 is the current density and the exchange current density, A/m^2, separately; F is Faraday's constant, C/mol.

The ohmic polarization V_{ohm} is calculated by:

$$V_{\text{ohm}} = i\frac{t_m}{\sigma_m} \tag{36}$$

where t_m is Proton exchange membrane thickness, m; σ_m is the conductivity of the membrane, S/m.

The concentration polarization V_{con} is calculated by Wang et al. (2021):

$$V_{\text{con}} = \frac{RT_{\text{el}}}{2\alpha F}\left[\ln\left(\frac{i_L}{i_L - i}\right)\right] \tag{37}$$

where i_L is the limiting current density, A/m^2.

For a fuel cell stack formed by connecting N fuel cells in series, the voltage V_{st} is expressed by:

$$V_{\text{st}} = NV_{\text{cell}} \tag{38}$$

The electricity generated by the fuel cell is:

$$E_g = 33.3 m_{H_2}\eta_{\text{FC}} \tag{39}$$

where η_{FC} is defined as the output electric energy divided by the input hydrogen energy of the fuel cell, m_{H_2} is the amount of hydrogen consumed, kg.

3. Optimization approaches for hybrid renewable energy systems based hydrogen storage

When considering the inherent randomness and volatility of renewable sources, the optimization of hybrid renewable energy systems (HRES) becomes essential to ensure stable and economical operation. Previous studies have demonstrated that HRES optimization spans several key dimensions, including initial site selection design, the incorporation of uncertainty models for stochastic optimization (Fan et al., 2025; Song et al., 2024), and the integrated sizing and co-planning of multiple energy carriers (Alabi et al., 2022). In practice, relying solely on a single renewable source such as solar or wind often necessitates large-scale deployment, which not only leads to considerable investment but also challenges system reliability.

The primary objective of Section 3 is to establish a comprehensive optimization framework for hybrid renewable energy systems (HRESs) with hydrogen storage. This section systematically outlines the constraints typically encountered in system planning, such as energy balance requirements, capacity limitations. It further defines the objective functions, encompassing technical, economic, environmental and social indices, thereby ensuring a multi-dimensional evaluation of system performance. In addition, various optimization algorithms, including classical, intelligent and hybrid approaches, are introduced, along with their implementation through software tools, which serve as a bridge between theoretical modeling and practical applications. By integrating these elements, the framework enables the effective utilization of the physical models presented in Section 2, facilitates the identification of optimal system configurations, and supports the reliable and cost-efficient deployment of hydrogen storage within HRESs.

3.1. Performance indicators for optimization of HRES

3.1.1. Technical index

Technical indicators refer to ensuring that the system responds to demand and maintains a stable operating state. Due to the characteristics of intermittent solar radiation and wind speed, the stability of HRES power generation is critical. A reliable power system needs enough electricity to meet the load demand in a certain period. The prevailing technical indicator utilized to assess the reliability of HRES is LPSP (Ayop et al., 2018). When the energy available from production and storage in batteries or other energy storage devices is insufficient to meet the load demand for hour "t", this deficit, termed loss of power supply (LPS) for hour "t", can be expressed as:

$$\text{LPS}(t) = E_L(t) - (E_G(t) + E_B(t-1) - E_{B_{\min}})\eta_{\text{inv}} \tag{40}$$

where $E_L(t)$ is load demand in hour t, kWh; η_{inv} is inverter efficiency; $E_G(t)$ is the energy generated by HRES for hour t, kWh; Specifically, the instantaneous power outputs of the photovoltaic and wind subsystems, as described in Eqs. (4) and (18), are integrated over the corresponding time interval to obtain their hourly energy contributions, which are then aggregated to calculate in the LPS formulation; $E_B(t-1)$ is energy stored in batteries in previous hour, kWh; $E_{B_{\min}}$ is the battery's minimum allowable energy level, kWh.

LPSP is the ratio of insufficient energy to the energy required by the load in a period of T:

$$\text{LPSP} = \frac{\sum_{t=1}^{T}\text{LPS}(t)}{\sum_{t=1}^{T}E_L(t)} \tag{41}$$

Similarly, another evaluated reliability indicator is expected energy not provided (EENS), which estimates the total hours within the study period (typically a year, with H = 8760) when demand surpasses available power generation capacity. EENS is calculated as

$$\text{EENS} = \sum_{k=1}^{8760} LD \tag{42}$$

where L is the average annual demand, kW; D is the duration (h) in which load is not met out.

Loss of energy expected (LOEE) is the expected energy not provided in a period of H and can be defined as

$$\text{LOEE} = \sum_{h=1}^{H}\sum_{i\in S}P_i\text{LOE}_i \tag{43}$$

where LOE_i is the energy loss (kWh) when HRES cannot satisfy the expected energy at hour h.

3.1.2. Economical index

The financial viability of an HRES is the primary factor in determining the size of the energy storage system and the overall design of the HRES. Economical assessment comprises initial investment, component replacement, operation and maintenance costs. At present, the commonly used economic indicators are net present cost (NPC), cost of energy (COE) and life cycle cost (LCC).

The NPC of an HRES system represents the present value of all installation and operational costs incurred over the system's lifetime. The COE, a crucial factor in developing a cost-effective hybrid system, indicates the average cost per kWh of useable electrical power generated by HRES (Sawle et al., 2018). COE can be expressed as follows:

$$\text{COE} = \frac{\text{NPC}}{\sum_{h=1}^{8760}P_L(t)}\text{CRF} \tag{44}$$

where P_L represents the hourly power consumption, kWh, and CRF represents the capital recovery factor. The CRF is calculated as:

$$\mathrm{CRF} = \frac{i_r(1+i_r)^r}{(1+i_r)^r - 1} \tag{45}$$

where i_r is the real annual interest rate, r is the lifetime cycle.

LCC is the cumulative cost from design to recovery of the system throughout its life cycle (Tezer et al., 2017).

$$\mathrm{LCC} = \sum_{i=1}^{n} (\mathrm{ICC}_i + \mathrm{RC}_i + \mathrm{OMC}_i) \tag{46}$$

where ICC_i is the installed capital cost ($/kW).

3.1.3. Environmental index

In deploying hydrogen-based hybrid renewable energy systems, CO_2 mitigation is a central concern, as these systems align with global initiatives to tackle climate change and accelerate the transition to cleaner, sustainable energy. Optimizing the components and operation of HES contributes directly to significant decarbonization across multiple sectors by displacing fossil-fuel-based systems. Therefore, the current Environmental Index was primarily established with a focus on greenhouse gas emissions, especially CO_2, given its paramount impact on climate change (Reza et al., 2026). The most commonly used environmental indicator is the one of greenhouse gas emissions (GHE), F_{GHE} (Wang et al., 2017).

$$F_{\mathrm{GHE}} = \sum_{i=1}^{n} N_i \mathrm{GHE}_i \tag{47}$$

where N_i is the number of the system component i, GHE_i is the greenhouse gas emission of a single component i.

However, focusing solely on greenhouse gas emissions cannot fully capture the environmental burdens of HRES. In addition to GHGs, acidification potential (AP) and eutrophication potential (EP) are also important impact categories. Acidification potential is usually expressed in terms of SO_2-equivalents (kg SO_2-eq), with contributions from SO_2, NOx, NH_3, HF, and HCl, based on internationally accepted stoichiometric approaches. Eutrophication potential, on the other hand, accounts for nutrient enrichment (e.g., nitrogen and phosphorus compounds) that can lead to excessive algae and plankton growth, deterioration of water quality, and loss of ecological value. Therefore, in this study, while F_{GHE} is used as a primary index, we acknowledge the necessity of extending the environmental evaluation framework to integrate AP and EP in future work to ensure a more comprehensive sustainability assessment (Kourkoumpas et al., 2018).

3.1.4. Social index

Social assessment examines the potential of the HRES to enhance the Human development index (HDI) through a reliable energy supply. It also considers societal aspects, including public acceptance of system deployment and the creation of employment opportunities (Sawle et al., 2018). The HDI contribution of the HRES can be expressed as:

$$\mathrm{HDI} = 0.0978 \ln \frac{P_{\mathrm{LOAD}} + \min\left(f_{\max,P_{\mathrm{SE}}} * P_{\mathrm{SE}}, f_{\max,P_{\mathrm{LOAD}}} * P_{\mathrm{LOAD}}\right)}{\eta_{\mathrm{HUMAN}}} - 0.0319 f_{\max,P_{\mathrm{SE}}} \tag{48}$$

where $f_{\max,P_{\mathrm{SE}}}$ is the maximum surplus energy that can be used for AC additional load; $f_{\max,P_{\mathrm{LOAD}}}$ is the element to increase yearly AC load; P_{SE} is yearly surplus energy, kWh/yr, η_{HUMAN} is the number of humans who consume the generated power.

Job creation can be expressed as (Adoum Abdoulaye et al., 2024):

$$\text{Job creation} = \sum_{S=1}^{S} \gamma_s \times \mathrm{P}_s \tag{49}$$

where P_s is the energy delivered by subsystem s, and γ_s is a job creation factor.

3.1.5. Constraints and design variables

In the optimization design, the objective function depends on the design variables, and the value ranges of these variables are subject to various restrictive conditions. Constraints are certain restrictions on the lower and upper bounds of design variables in the optimization process and on the performance parameters that are not considered in the objective function. Common constraints for optimization of HRES are mainly divided into eight aspects, including:

- Number of units (Number of PV panels N_{PV}, wind turbines N_{WT}, batteries N_{BAT}, fuel cells N_{FC}, H_2 tanks N_{TANK}, inverter N_{INV}, electrolysers N_{ELE}, diesel generators N_{DG})
- Renewable capacity (Insolation, wind capacity)
- Installation sizes (Area covered by PV modules A_{PV}, area swept by wind turbine blades A_{WT})
- Physical constraints (PV panel tilt angle i_{PV}, wind turbine installation height H_{W})
- Energy storage device constraints (SOC of battery, charge/discharge rate, cycle life, hydrogen storage tank capacity C_{tank}, Battery bank storage capacity C_{BAT})
- Economic constraints (Fixed budget)
- Policy constraints: To reflect national or regional regulations and incentives, design variables are additionally constrained by policy-driven upper and lower limits (Hossain Lipu et al., 2025). For example, maximum or minimum capacities of electrolysers, fuel cells or H_2 storage tanks can be set according to subsidies, tax incentives, or regulatory targets. These constraints ensure that the optimized HRES configuration is compliant with local hydrogen deployment strategies and incentive programs (Reza et al., 2026).
- Other technical constraints as applicable, e.g., grid interconnection limits, voltage or frequency requirements.

Optimizing the HRES can obtain the best results from limited energy resources and investments. The design variable refers to the quantities to be determined and is related to the constraint condition and the objective function involved in the optimization problem. The design variables of an HRES include the size specifications of each component or a set of possible configurations (for example, the selection of the type of microgrid bus, wind turbines i_{WT}, battery banks i_{BAT}, diesel generators i_{DG}).

3.2. Optimization techniques and algorithms

The optimal methodologies mainly include classical algorithms, intelligent techniques, hybrid techniques, and software tools. Classical approaches often employ calculus to generate optimal solutions. Intelligent and hybrid techniques have become very popular in solving complex optimization issues. These approaches can achieve global system optimization, thereby improving convergence and accuracy in the search for optimal solutions.

3.2.1. Classical algorithms techniques for optimization

Optimization relying on traditional methods is generally deterministic, in which mathematical models are applied to search for the global optimum. One advantage of this approach lies in its ability to yield a precise solution. However, it is often inadequate when dealing with a high-dimensional and complex decision space. At the preliminary stage of design and optimization, conventional strategies usually adopt numerical, probabilistic, iterative algorithms and graphical methods (Thirunavukkarasu et al., 2023).

Numerical methods refer to approaches that obtain approximate rather than exact solutions through mathematical procedures and computational techniques. In the optimization of hydrogen-based hybrid renewable energy systems (HRES), numerical methods are widely employed, including linear programming (LP), nonlinear programming (NLP), mixed-integer linear programming (MILP), and dynamic programming (DP) (Emrani and Berrada, 2024). These approaches optimize the sizing of components, such as solar photovoltaics, wind turbines, fuel cells, electrolyzers, and energy storage systems, to minimize energy loss, reduce system costs, and ensure a stable power supply. In HRES, LP can be used to optimize the dispatch of renewable energy and storage systems by modeling the interactions between various components. However, its application is limited to cases where linear functions accurately describe the relationships within the integrated systems (Alabi et al., 2022). Mixed-integer linear programming (MILP) extends LP to handle more complex relationships between variables, which are common in energy systems due to the nonlinear behavior of power generation and consumption (Gao et al., 2022). Because one of the system size limitations is that the number of system configurations must all be integers, MILP is a relatively common mathematical method for optimizing hybrid energy systems. For example, considering the load demand and energy supply potential, the MILP method is used to solve the position of the generators and the design of the microgrid (Ferrer-Martí et al., 2013). Amrollahi and Bathaee (2017) used demand response processes and MILP algorithms to optimize the size of microgrid components. Dynamic programming (DP) is well-suited for time-varying optimization problems, such as managing energy storage or scheduling energy resources based on predicted renewable energy availability and demand. DP can account for the stochastic nature of HRES by incorporating future uncertainties in energy generation and demand profiles. Hydrogen storage and flow control are crucial for hydrogen-based power systems. Dynamic programming-based hydrogen storage control systems can obtain global optimal points.

Graphical methods are effective for optimization problems with two variables, but become impractical for high-dimensional cases, as they rely on simplified estimations. Despite their ease of use, they are limited in handling complex systems. In energy system optimization, the probabilistic method models uncertainties in system components through probability distributions, enabling decision-making under variable conditions. By representing uncertainty and random variations of system components through probability distributions, it supports informed decision-making under uncertain conditions. This method requires relatively limited data to yield dependable results, making it suitable when information is scarce. Its main limitation, however, lies in the restricted number of variables it can accommodate, typically one or two, which reduces accuracy in complex systems (Reza et al., 2026).

In addition to the classical optimization algorithms mentioned above, iterative algorithms are also commonly used to optimize the sizing of HRES with hydrogen storage (Elkholy et al., 2024). Smaoui et al. (2015) developed an iterative optimization algorithm for the optimal sizing of a stand-alone hybrid PV/wind system with hydrogen storage, ensuring reliable energy supply to a desalination unit while minimizing installation costs and demonstrating the technical feasibility of hydrogen-based storage solutions. Although iterative algorithms are efficient and easy to implement, they take a considerable amount of time due to the complex calculations that require multiple loops. Additionally, classical algorithms frequently encounter limitations when addressing non-convex, high-dimensional, and multi-objective problems, which are prevalent in HRES (Saxena et al., 2025). Table 2 summarizes the characteristics of classical algorithms commonly used to optimize hydrogen-based HRES. The dependency on accurate mathematical models and the difficulty in handling uncertainties restrict the scope of classical algorithms in dynamic and highly nonlinear systems.

3.2.2. Intelligent algorithms techniques for optimization

In hydrogen-integrated hybrid renewable energy systems (HRESs), the complexity arising from integrating multiple renewable sources with advanced storage solutions necessitates the adoption of sophisticated optimization strategies. Traditional optimization methods, although effective for simple or convex problems, are often limited by inflexibility, computational inefficiency, and difficulty in handling multi-objective, stochastic, or non-linear scenarios (Saxena et al., 2025). Artificial intelligence (AI) offers adaptive, flexible, and efficient alternatives, transforming mathematical models into practical decision-making processes capable of addressing the inherent intermittency, uncertainty, and multi-objective trade-offs of HRESs. Intelligent optimization algorithms for HRESs can be classified into three broad categories: non-swarm biological heuristics, swarm-based biological heuristics, and physical/chemical-inspired heuristics.

Non-swarm biological heuristics, such as genetic algorithms (GA), whale optimization algorithm (WOA), and biogeography-based optimization (BBO), mimic evolutionary or biological processes without relying on swarm intelligence. GA is the most widely used method for HRES optimization due to its robust exploration of complex, high-dimensional solution spaces via selection, crossover, and mutation (Agoundedemba et al., 2025). Advanced variants such as NSGA-II and NSGA-III allow multi-objective optimization (MOO), balancing conflicting goals such as minimizing cost while maximizing system reliability. Limitations include potential premature convergence and sensitivity to parameter settings.

Swarm-based heuristics, including particle swarm optimization (PSO), firefly algorithm (FA), and ant colony optimization (ACO), emulate collective behaviors in nature. PSO is particularly effective for real-time energy dispatch and hydrogen management, leveraging particle interactions to converge on high-quality solutions. Modified versions, such as MOPSO or adaptive PSO variants, enable multi-objective optimization, addressing the challenge of balancing hydrogen production, storage, and fluctuations in renewable generation (Cheraghi and Hossein Jahangir, 2023). While swarm methods excel at convergence speed, they may struggle to escape local optima and require careful parameter tuning.

Physical or chemical heuristics, such as simulated annealing (SA), harmony search (HS), and Big Bang-Big Crunch (BB-BC), are inspired by physical and thermodynamic processes. SA, for example, is suitable for global search in large solution spaces, while HS offers discrete and continuous optimization capabilities. These algorithms are particularly useful when problem landscapes are rugged or highly constrained, but often converge more slowly than swarm or evolutionary methods.

Hybrid algorithms combine two or more techniques to leverage complementary strengths. For example, GA-PSO hybrids exploit GA's exploratory capability and PSO's rapid convergence, achieving superior cost-effectiveness and reliability in HRES design (Bade and Tomomewo, 2024). Similarly, SA-TS or CS-HS-SA-ANN hybrids integrate global and local search strategies, thereby improving convergence efficiency and system-level performance (Zhang et al., 2019). Recent comparative studies further highlight the performance differences among these

Table 2
Classical methods and their advantages and disadvantages (Saxena et al., 2025).

Optimization Technique	Computational Efficiency and Simplicity	Longer Computation Time	Simplified Sizing Methodologies	Visual Optimization Solutions	Increased Computational Efforts
Numerical	✓	✓			
Probabilistic			✓		
Graphical				✓	
Iterative		✓			✓

algorithms. Talebi and Aly (2025) reported that PSO consistently achieved the lowest NPC for both Li-ion and lead-acid battery-based microgrids, owing to its rapid convergence and efficient search capability. FA ranked second, showing strong exploration ability and good economic performance, while GA delivered stable yet moderate results with slower convergence. ACO performed the worst, with higher NPC, poor convergence, and limited suitability for cost-sensitive applications. Based on these insights, PSO and FA are often selected as core components for hybrid models. For instance, the FA-PSO hybrid first uses FA to broadly explore the search space and then applies PSO for rapid refinement, achieving lower NPC and greater renewable energy penetration than single algorithms. Table 3 summarizes the key features, advantages, and limitations of each algorithm category, providing a framework for selecting appropriate methods depending on system characteristics, objectives, and computational resources.

3.3. Software tools of optimization and their applications

The optimization of HRES that incorporates hydrogen storage requires advanced software tools to model, simulate, and optimize system performance under variable and often unpredictable conditions. Several software platforms are specifically designed to address the complex, multi-objective optimization challenges of HRES, enabling the efficient integration of renewable energy sources with hydrogen storage and other components. The most widely used tools in this field include HOMER, MATLAB/Simulink, iHOGA, TRNSYS and RETScreen. Each tool offers unique capabilities suited to different aspects of HRES design, analysis, and operation, contributing to improved system efficiency, cost savings, and reliability (Anoune et al., 2018). Table 4 provides a comprehensive analysis of each software, highlighting its respective advantages and disadvantages.

Among the available software platforms, HOMER remains one of the most widely applied tools for HRES optimization, providing a comprehensive framework for system modeling, techno-economic evaluation, and sizing. It enables users to assess different combinations of renewable energy sources, hydrogen storage units and operational strategies for both off-grid and grid-connected applications (Hussam et al., 2024). Moreover, HOMER software can be used for sizing optimization of off-grid and grid-connected systems. Based on optimization results, HOMER software can perform further sensitivity analyses, such as generator minimum load, battery SOC setpoint, and changes in PV panel and wind turbine technologies. However, the software can only achieve SOO by minimizing NPC (Al-falahi et al., 2017).

MATLAB/Simulink is another powerful tool widely used for HRES optimization, offering flexibility through its extensive library of toolboxes and advanced programming capabilities. MATLAB/Simulink enables the modeling, simulation, and optimization of complex HRES configurations by supporting deterministic and stochastic approaches. Its Optimization Toolbox enables the implementation of various optimization algorithms, including GA, PSO, and hybrid approaches specifically tailored to HRES applications. MATLAB is especially well-suited for integrating intelligent algorithms, such as Neural Networks and Fuzzy Logic, which enhance real-time decision-making and predictive

Table 4
Comparison of the various software tools for HRES.

Software	Advantages	Disadvantages
HOMER	• Analyze stand-alone and on-grid systems (Deshmukh, 2019) • Analyze a system of multiple energy mixtures that include wind, solar, biomass and water power (Tozzi and Jo, 2017) • Conduct a comprehensive analysis, including technical and economic aspects (Bansal et al., 2020)	• Cannot analyze thermal system • Regardless of the depth of discharge of the battery (Ram et al., 2021) • Only a single objective function is allowed to minimize NPC (Sinha and Chandel, 2014)
MATLAB/Simulink	• A detailed physical model can be established through mathematical models (Chong et al., 2017) • Optimized through MATLAB using a variety of optimization algorithms	• Compared with other software, the model established by Simulink is more complicated (Sharma et al., 2019)
iHOGA	• In-depth optimization is feasible in iHOGA (Ram et al., 2021) • Multi-objective optimization is feasible (Zubi et al., 2016)	• Not compatible with systems updated after Windows XP−More complex (Sinha and Chandel, 2014) • The model configuration it builds has certain limitations (Ram et al., 2021)
TRNSYS	• A wide range of applications and highly flexible models (Sauter, 2017) • Real system performance can be simulated without many details (Ram et al., 2021) • Mixed with MATLAB	• Inability to conduct economic and environmental analysis (Bakić et al., 2012) • Cannot analyze systems containing biomass energy and hydropower (Tozzi and Jo, 2017)
RETScreen	• Deals with both electrical and thermal renewable systems • An adequate tool for pre-feasibility study at the conceptual design phase (Lee et al., 2012)	• No methods or functions for optimization • Generator, hydro energy, bio energy, and thermal system analysis cannot be performed

Table 3
Comparison of intelligent optimization algorithms for hydrogen-integrated HRES.

Algorithm Category	Key Examples	Main Characteristics	Advantages	Limitations	Typical Applications
Non-swarm Biological Heuristics	GA, NSGA-II, NSGA-III, WOA, BBO	Mimic natural evolution; population-based; selection, crossover, mutation mechanisms	Strong global search; robust in high-dimensional, multi-objective optimization; well-studied	May converge prematurely; sensitive to parameter tuning; moderate computational cost	Component sizing, energy dispatch, storage optimization in HRES; multi-objective trade-offs (cost vs reliability)
Swarm-based Biological Heuristics	PSO, MOPSO, FA, ACO, ABSO	Inspired by collective behaviors of animals/insects; particles or agents communicate and adapt	Fast convergence; dynamic adaptation to real-time energy fluctuations; suitable for multi-agent systems	Risk of local optima; parameter tuning required; traditional PSO is single-objective	Real-time dispatch, hydrogen production scheduling, renewable-storage coordination; MOO with modified variants
Physical/ Chemical Heuristics	SA, HS, BB-BC, MBA	Inspired by physical/chemical phenomena (annealing, harmony search, cosmology); iterative global search	Effective for complex, constrained, or rugged solution spaces; easy to implement	Slower convergence; may require hybridization for large-scale, multi-objective problems	Optimization of system configuration, overall energy management; discrete and continuous HRES design problems
Hybrid Algorithms	GA-PSO, SA-TS, CS-HS-SA-ANN	Combine two or more algorithmic principles; exploit complementary strengths	Improved convergence speed and solution quality; better handling of multi-objective and dynamic constraints; robust	Increased complexity; higher computational requirements; design and tuning can be challenging	Comprehensive HRES optimization: component sizing, energy dispatch, hydrogen storage utilization; multi-objective techno-economic-environmental optimization

control in HRES. The ability to run simulations with real-time data inputs also makes MATLAB/Simulink ideal for designing and testing control strategies for hydrogen storage, ensuring stable and cost-effective operation under fluctuating energy conditions.

iHOGA is a software tool specifically designed to optimize HRES using GA. Developed to model and simulate hybrid systems with an emphasis on economic feasibility, iHOGA excels in off-grid and micro-grid applications. It enables users to determine the optimal combination of renewable sources and storage capacities, such as hydrogen storage, by balancing cost, reliability, and environmental impact. iHOGA's optimization engine supports multi-objective optimization, ensuring the system configuration achieves minimal cost while satisfying energy demands and emission-reduction goals. Compared to other HRES optimization software, the iHOGA software determines the optimal system configuration and enables designers to assess the economic and technical viability of various technologies, including NPC, CO_2 emissions, and unmet load (Shamachurn, 2021).

TRNSYS is a versatile software package that models dynamic energy systems, providing detailed simulations of HRES with both short-term and long-term energy storage. TRNSYS enables the integration of complex components, such as electrolyzers, fuel cells, and storage systems, making it particularly useful for studying the impact of hydrogen storage on system stability, energy balance, and operational efficiency. It also supports component-level customization, enabling detailed control over energy flows within HRES, which is essential for systems incorporating hydrogen as both a storage and fuel option. The author of this study has conducted case studies based on the local weather conditions in Yichang, Hubei, China (Li et al., 2022). The performance of a PV/WT/FC/diesel engine generator system based on HRES is simulated by TRNSYS software and compared with the one-year load situation of a local community. TRNSYS can interconnect any system component appropriately and solve the relevant differential equations for each connection and system; however, the software lacks optimization settings and only offers parametric studies. To solve this problem, a neural network genetic algorithm or the TrnOpt component in TRNSYS can realize the optimal design of HRES. The LPSP of the HRES with solar/wind and hydrogen storage system was simulated by TRNSYS. The results show that the hydrogen storage method enhances the system's reliability and reduces reliance on grid electricity. Following that, using an ANN-GA approach, the optimal design has the lowest installation cost, CO_2 production and LPSP (Izadi et al., 2022).

RETScreen is a comprehensive energy management and decision-support tool developed by Natural Resources Canada. It assesses the technical and financial viability of energy projects, including renewable energy and energy-efficient technologies. RETScreen offers a broad range of features for HRES feasibility analysis, allowing users to simulate system configurations, assess energy production potential and conduct financial analysis to determine project viability. RETScreen's cost-benefit analysis features help optimize the economic aspects of HRES with hydrogen storage, while its environmental assessment capabilities enable the evaluation of greenhouse gas reduction benefits. Additionally, RETScreen provides tools for risk analysis and sensitivity testing, which are valuable for identifying and mitigating risks associated with integrating hydrogen storage in HRES configurations (Khalil and Dincer, 2023).

Table 5
Summary of commonly used optimization methods.

Optimization method	Advantages	Disadvantages	Tools
Classical	• Mature • Definite answer (Memon and Patel, 2021)	• Many iterations • Cannot handle a large number of variables	MILP, DIRECT, graph theory
Intelligent algorithms	• Higher accuracy • Good convergence (Al-falahi et al., 2017)	• Complex • Each execution different	GA, PSO, NSGA-II, MOGA, MOGA-II
Hybrid algorithm	• Good robustness (Memon and Patel, 2021)	• High computing power • More complex	SA-ANN, HS-ANN, HCHSA
Software tools	• Accurate physical model	• Limited optimization	HOMER, MATLAB, RETScreen, iHOGA

3.4. Summary and evaluation

As shown in Table 5, a wide range of optimization strategies for HRES component sizing—including classical, intelligent, and hybrid techniques, as well as software-based approaches—have been explored. Classical optimization methods, such as MILP and graph-theoretical approaches, are computationally efficient and yield deterministic results, making them well suited for small-scale or single-objective problems. Nevertheless, their limited flexibility in handling complex, multi-objective, and uncertain scenarios becomes a major limitation when multiple nonlinear objectives and stochastic parameters are involved, such as fluctuating hydrogen demand, intermittent renewable generation, or dynamic storage efficiency. These methods typically require linearizing or simplifying constraints, which may yield suboptimal or unrealistic solutions when applied to large-scale hydrogen-integrated HRES.

Intelligent algorithms, particularly GA, PSO, and NSGA-II, have therefore gained popularity due to their adaptability and robustness in tackling multi-objective optimization problems. They can efficiently explore complex solution spaces and incorporate stochastic uncertainties, making them suitable for scenarios with conflicting objectives, such as cost, environmental impact, and safety considerations. However, their performance depends heavily on parameter tuning, and they may suffer from non-repeatability across runs or premature convergence, leading to suboptimal solutions.

Hybrid algorithms have emerged as a promising paradigm that combines the complementary strengths of classical and intelligent methods. For instance, SA–ANN or HS–ANN frameworks integrate the global search capabilities of metaheuristics with the local learning capacity of neural networks, enhancing both convergence speed and solution accuracy. These approaches are particularly well-suited for large-scale hydrogen-based systems, where multiple coupled variables—such as energy generation, storage, and hydrogen transport—must be simultaneously optimized. Nevertheless, the high computational cost and complexity of parameter calibration remain challenges for practical implementation.

Given the increasing complexity of HRES planning, future research should prioritize the development of hybrid or adaptive optimization frameworks that can seamlessly integrate multiple objectives, stochastic uncertainties, and safety constraints. In particular, algorithms that incorporate predictive modeling, real-time adaptability, and uncertainty quantification will be crucial for achieving reliable and socially acceptable hydrogen-integrated energy systems.

Beyond optimization theory, hydrogen storage and safety considerations are central challenges in HRES design, but are often oversimplified in current optimization frameworks. Hydrogen storage technologies—ranging from compressed gas and liquid hydrogen to solid-state storage—differ significantly in energy density, leakage behavior, and safety risks. For instance, compressed hydrogen offers high reversibility but requires high-pressure vessels (up to 700 bar), increasing explosion hazards and maintenance costs. Liquid hydrogen offers a higher volumetric density but requires cryogenic conditions and suffers from boil-off losses. Solid-state storage, though safer, still faces slow kinetics and high material costs. These trade-offs underscore that storage technology cannot be optimized solely on energy or cost grounds; instead, safety, reliability, and lifecycle impacts must be integrated into the optimization objectives. Intelligent and hybrid algorithms offer an opportunity to incorporate probabilistic safety metrics —such as failure rates, leakage probability, and dispersion modeling —into multi-objective frameworks, thereby aligning system design with real-world safety requirements.

Policy and economic dimensions also play a critical role in shaping the optimization landscape, but are rarely systematically embedded in current models. Policy-driven incentives, such as carbon pricing, renewable energy credits, and safety standards, directly influence cost structures and the feasibility of technologies. For instance, integrating feed-in tariffs or hydrogen tax credits into techno-economic models can fundamentally shift the optimal configuration of an HRES (Okonkwo et al., 2025). Moreover, local safety regulations and hydrogen codes (e.g., ISO, 19880) impose spatial and operational constraints that affect siting and design. Therefore, next-generation optimization frameworks should couple techno-economic and environmental objectives with policy and social constraints, enabling dynamic decision-making that reflects both market mechanisms and regulatory compliance.

Finally, while software tools such as HOMER, MATLAB, RETScreen, and iHOGA have greatly facilitated energy system simulation, their current implementations remain limited by model simplification and weak coupling between physical and safety processes. The integration of artificial intelligence and digital twin technologies represents a promising direction—allowing models to evolve from static optimization toward adaptive, data-driven platforms capable of real-time monitoring, uncertainty quantification, and predictive safety management. This convergence of optimization algorithms, hydrogen safety modeling, and policy-aware decision frameworks will be essential for the next stage of hydrogen-integrated HRES research, ensuring systems that are not only economically and environmentally optimized but also safe, resilient, and socially acceptable.

4. Artificial intelligence for simulation and optimization of hybrid renewable energy systems

Artificial intelligence (AI) techniques enable the accurate prediction of the performance and efficiency of HRES. These techniques enhance energy yield and minimize energy losses under various environmental parameters, including temperature, sunlight exposure, wind speed and humidity. This section focuses on artificial intelligence for predicting and optimizing HRES.

4.1. Artificial intelligence for prediction of hybrid renewable energy systems

Artificial intelligence (AI) includes evolutionary computation, expert systems, fuzzy logic, machine learning (ML), computer vision, recommendation systems, etc. ML is the most widely used AI method for renewable energy systems. ML, which uses a large amount of data or previous experience to train algorithms, is the core of AI and the necessary way to make computers intelligent. After a long period of development, its traditional research directions, including artificial neural networks (ANN), support vector machines (SVM), decision trees (DT), random forests (RF), Bayesian learning, and other applications, have become relatively mature (Wang et al., 2021). Deep learning (DL) has been rapidly developed as an extension of machine learning in recent years, utilizing deep neural networks. Compared with machine learning, deep learning can extract more input features independently and is closer to the goal of analyzing and thinking like the human brain (Ahmad et al., 2021). The relations among artificial intelligence, machine learning and deep learning are shown in Fig. 3.

Artificial intelligence (AI) techniques have become essential for improving prediction accuracy and decision-making efficiency in hybrid renewable energy systems (HRESs), which comprise multiple stochastic components, such as solar and wind generation, hydrogen production, and storage, and exhibit pronounced temporal and nonlinear interactions. As a result, machine learning (ML) and deep learning (DL) models are increasingly employed to forecast critical variables, including renewable resource availability, system output, energy demand, and electricity prices. These forecasts serve as the foundation for system design, energy management, and operational optimization, directly informing hydrogen production scheduling and storage-dispatch strategies. Fig. 4 presents an overview of machine learning algorithms commonly used for predicting renewable energy systems.

Initial work in this field focused on individual prediction tasks, such as solar irradiance and wind speed forecasting, using conventional approaches such as support vector machines (SVMs) and feed-forward neural networks (ANNs). For example, Nabavi et al. (2021) and Rangel-Martinez et al. (2021) demonstrated the viability of ANN-based models for meteorological forecasting, yet these methods were often constrained by reliance on fixed input features and limited ability to capture spatio-temporal correlations. In response, recent advances have moved toward hybrid and ensemble frameworks that integrate signal decomposition, heuristic optimization and deep learning techniques. Zhu et al. (2024) for instance, proposed a hybrid MEMD–ALO–BiLSTM model for solar irradiance forecasting, effectively addressing signal noise and non-stationarity. Similar hybrid strategies—such as combining CNN feature extraction with LSTM temporal modeling—have been applied in wind power and hydrogen output prediction, demonstrating improved adaptability to increasingly dynamic operating conditions.

As the field evolves, research has shifted toward system-level forecasting that encompasses not only renewable generation but also power output, load demand and market signals. Benitez and Singh (2025) pointed out that hybrid deep-learning architectures outperform traditional ML when predicting energy output under fluctuating environmental conditions—an advancement particularly relevant to hydrogen-integrated HRES, where forecast accuracy of renewable generation directly informs electrolyzer and storage operation. In parallel, load and electricity-price forecasting has progressed from isolated models to multi-task and co-learning architectures. Heistrene et al. (2023), for example, applied ensemble and explainable AI methods in dynamic price prediction, enabling interpretable, data-driven dispatch decisions.

4.2. Artificial intelligence-based optimization of hybrid renewable energy systems

In recent years, machine learning (ML) has become increasingly integral to optimizing energy systems, both in industrial applications and academic research. In HRES with hydrogen storage, ML is utilized for tasks such as time-series forecasting, classification, and regression analysis to predict energy generation, demand, and storage levels. This data-driven approach enables real-time management, allowing HRES to anticipate and adjust to variable energy flows from renewable sources. Meanwhile, optimization techniques, rooted in mathematical modeling, address planning challenges like capacity allocation, scheduling and energy dispatch. By defining objective functions and constraints, optimization algorithms guide decision-making on energy allocation, aiming to minimize costs or maximize system performance.

Optimization algorithms have traditionally been a go-to solution for engineering problems, but they are computationally intensive and may require significant processing resources, which can limit real-time applications. Computational load is influenced by hardware capacity (such as processing speed and memory) as well as the problem's complexity and structure, whether it involves linear or non-linear formulations, integer variables, or multi-dimensional constraints. When renewable systems face dynamic operational demands, the time required for optimization algorithms to reach a solution can be challenging. However, with the growing availability of historical data and high-speed computing, ML-based models have become a valuable alternative, offering faster forecasting for energy generation and storage needs. Still,

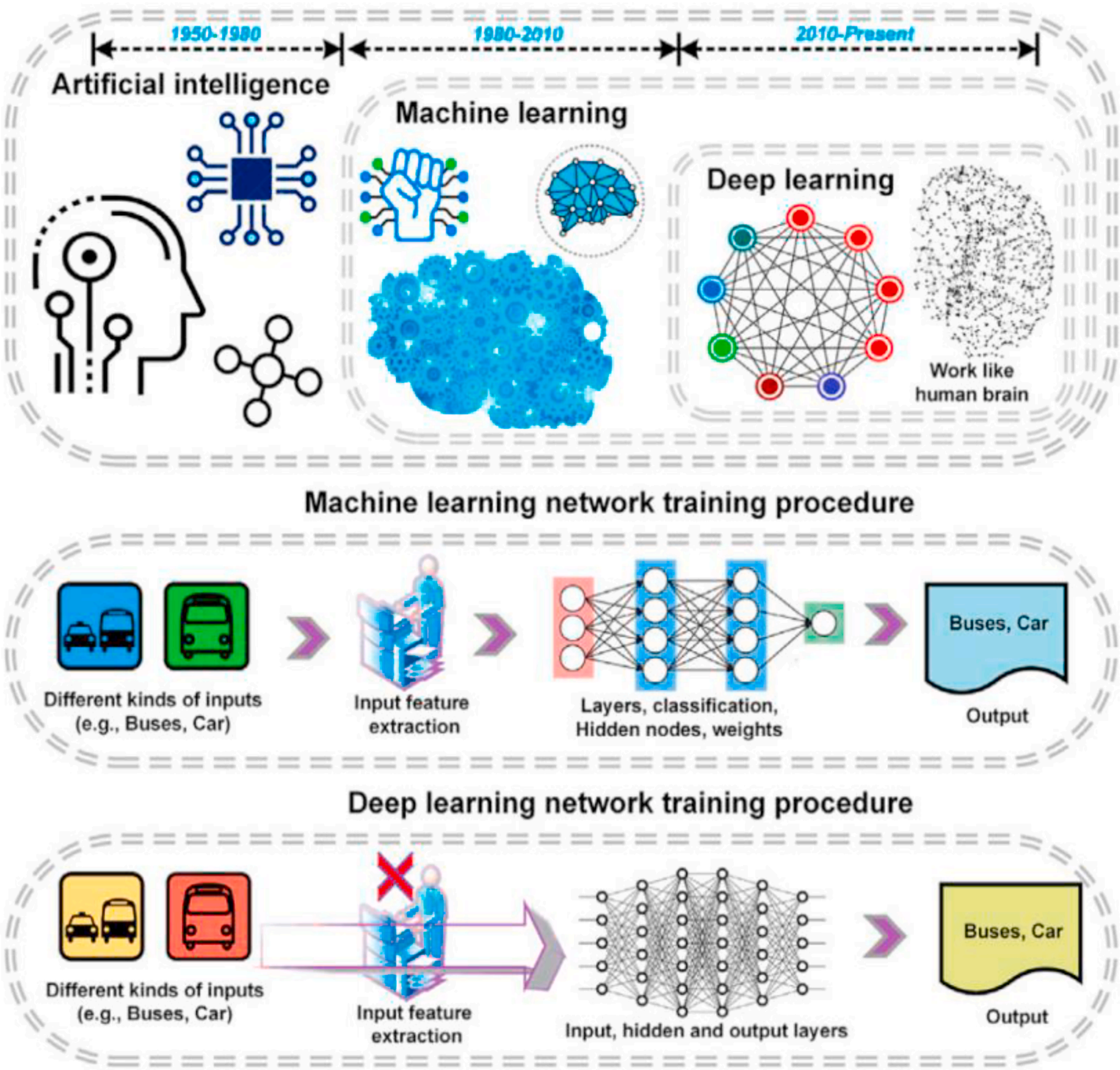

Fig. 3. Relations among AI, ML and DL and training procedures of neural networks (Ahmad et al., 2021).

ML techniques have limitations; for instance, many ML models are "black-box" algorithms, which lack transparency in decision-making processes and don't inherently offer optimal planning capabilities (Alabi et al., 2022). Recognizing these strengths and weaknesses, researchers are increasingly combining ML and optimization methods to capitalize on their complementary advantages. This integrated approach offers promising solutions for HRES, particularly as both fields continue to evolve with more sophisticated algorithms and computational capabilities.

4.2.1. Hyperparameter optimization in machine learning for enhanced accuracy

The accuracy and robustness of ML models are influenced significantly by hyperparameters, such as the number of hidden layers in neural networks, the selection of activation functions, learning rates, and training algorithms. Optimization methods have been widely adopted for fine-tuning these hyperparameters to improve ML model performance. By framing hyperparameter selection as an optimization problem where minimizing prediction error serves as the objective function, the structure of the machine learning algorithm represents the constraints, and hyperparameters are treated as decision variables, researchers can achieve a globally optimal combination of hyperparameters for model training. For example, GA has been used to optimize hyperparameters in ANNs in our previous works (Li et al., 2024). In those cases, GA iteratively searches for optimal weights and biases to minimize ANN prediction errors. Similarly, PSO has been employed to tune the parameters of convolutional neural networks (CNNs) and LSTM models for energy consumption forecasting, achieving high accuracy and generalization capability through this optimization-enhanced machine learning approach (Kim and Cho, 2019). Unlike PSO, which is prone to getting stuck in local optima due to limited diversification, and GA, which requires many evaluations to reach global optimality, GWO offers a balanced approach between exploration and exploitation through its multiple elite leaders and adaptive search mechanisms, ensuring computational efficiency and diversity across various optimization tasks. Therefore, as shown in Fig. 5, Xie et al. (2020) present an enhanced GWO-based evolving CNN-LSTM model for time series forecasting, where each wolf represents a set of network topology and learning hyperparameters, demonstrating improved performance by overcoming local optima stagnation and slow convergence through novel search strategies. These integrated approaches significantly enhance the performance of ML models, providing robust solutions for renewable energy predictions within HRES. This integration of optimization with ML model training not only

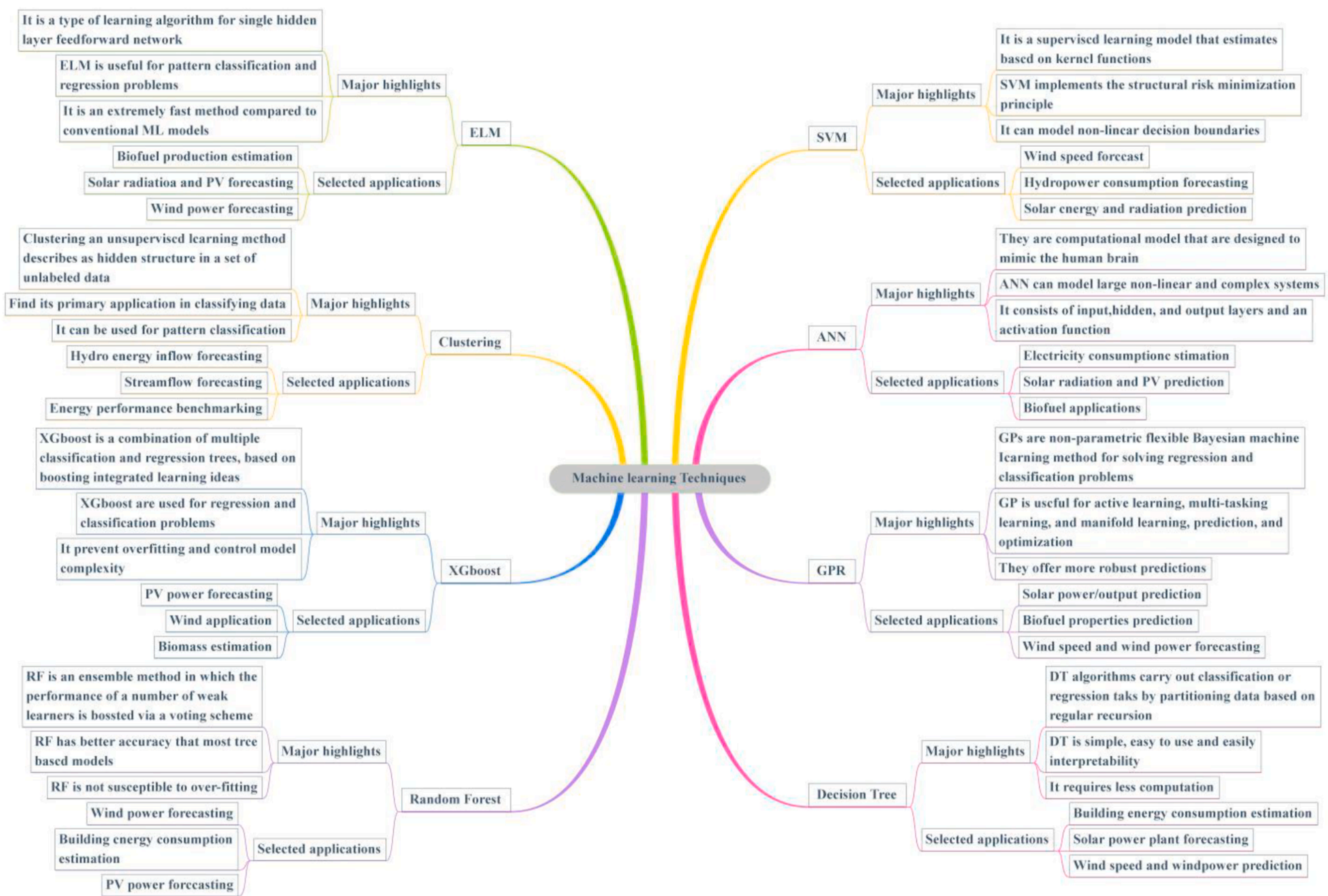

Fig. 4. Machine learning algorithm models for renewable energy applications (Alabi et al., 2022).

increases prediction accuracy but also enhances model robustness, enabling HRES to rely on more accurate forecasts of renewable energy and demand. As renewable generation can vary significantly, particularly with solar and wind sources, accurate predictions are crucial for balancing generation, storage, and energy dispatch within HRES.

4.2.2. Integrated prediction and optimization for uncertainty modeling and decision-making

Uncertainty in energy generation and demand is a defining challenge for HRES, particularly for those relying on intermittent renewable sources, such as solar and wind. Accurate and timely management of these uncertainties is essential for efficient energy dispatch, optimal hydrogen storage utilization and system reliability. The integration of machine learning (ML) with prediction and optimization techniques for decision-making provides a powerful solution, as these approaches complement each other in managing complex and dynamic energy systems.

Artificial neural networks (ANNs) provide a data-driven approach to modeling nonlinear relationships within HRES, making them invaluable for forecasting, system control, and energy management. In HRES with hydrogen storage, ANNs can analyze historical data to predict renewable energy generation and load demand, informing optimal dispatch and storage strategies. ANNs excel at pattern recognition, enabling them to capture the temporal and spatial variability of renewable resources, which is critical for proactive energy management in HRES. When integrated with optimization techniques such as GA or PSO, ANNs enhance the system's predictive capabilities, allowing for anticipatory control of hydrogen production and storage. This predictive control is particularly beneficial in HRES, as it enables better utilization of available resources, minimizes energy shortages and reduces operating costs by optimizing storage levels in anticipation of fluctuating renewable energy inputs. For example, Izadi et al. (2022) explore an HRES designed for zero-energy buildings in various climates, using an ANN model to predict key metrics like LPSP, cost rate and CO_2 emissions, while a GA optimization model identifies the most efficient configuration of PV panels, WT and hydrogen storage. This ANN-GA approach ensures the HRES achieves minimum installation cost, reduced CO_2 emissions and high system reliability, customized for each building's specific energy demands and local climate conditions. Yilmaz and Turksoy (2023) present an ANN-based control method for managing active and reactive power in a grid-connected hydrogen fuel cell system, achieving high power factor, low harmonic distortion and robust control, with ANN accuracy exceeding 97% across varied operational scenarios.

Reinforcement learning (RL) also plays a critical role in real-time optimization by enabling systems to continuously learn and adapt from interactions within dynamic environments, thereby improving decision-making policies over time (Ren et al., 2024). In hydrogen-based HRES, RL is particularly advantageous as it effectively manages the complexity of integrating multiple, intermittent energy sources alongside hydrogen storage and consumption requirements. For example, the deep deterministic policy gradient (DDPG) algorithm, a type of deep reinforcement learning, has been effectively applied to optimize energy scheduling in off-grid large-scale hydrogen production systems. By learning from real-time system feedback, the DDPG-based approach can dynamically adjust energy allocation to balance the inputs of renewable energy, hydrogen production, battery storage, and load demands, ultimately enhancing economic efficiency and ensuring operational safety (Liang et al., 2024).

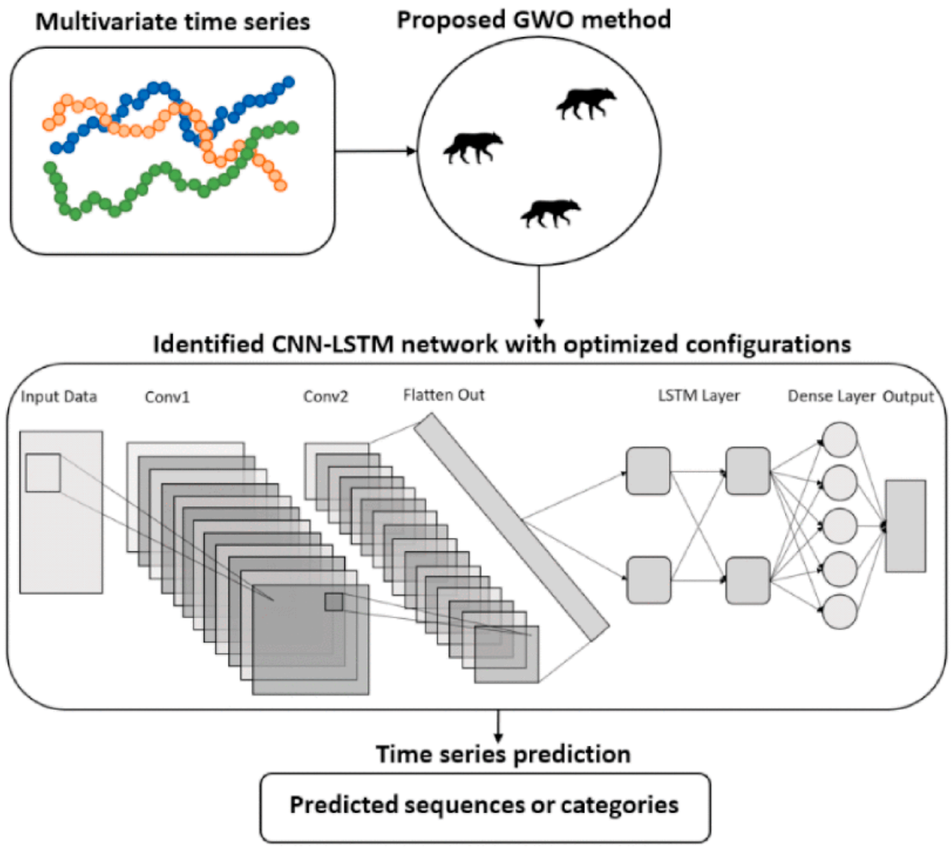

Fig. 5. The structure of the GWO-based evolving CNN-LSTM time series forecasting model (Xie et al., 2020).

4.3. Challenges and future directions

While the integration of machine learning (ML) and optimization holds significant promise for enhancing hybrid renewable energy systems (HRES) with hydrogen storage, several critical challenges remain to be systematically addressed to realize their full potential. One major limitation is the dependence of ML models on large volumes of high-quality, domain-specific datasets. Reliable model training requires data with adequate temporal resolution and coverage, encompassing variables such as renewable generation, hydrogen production and consumption, storage pressure, and system response under various operating conditions. However, in most HRES applications, such comprehensive datasets are rarely available due to the high cost and technical difficulty of monitoring distributed subsystems. Moreover, inconsistencies in data acquisition, arising from sensor drift, communication delays or data loss, can introduce noise and bias, ultimately degrading the accuracy and robustness of predictive models. To overcome these limitations, data augmentation techniques, physics-constrained data synthesis, and transfer learning approaches have recently gained attention, allowing models trained on one site or system configuration to be adapted to others with minimal retraining effort.

Beyond data availability, the computational complexity of real-time optimization is another barrier to practical deployment. Large-scale HRES typically involves coupled nonlinear dynamics among electrolyzers, fuel cells, hydrogen tanks, and renewable generation units. Traditional optimization methods, when integrated with ML-based forecasting, can become computationally intensive, especially when solving mixed-integer or stochastic formulations in real-time. Recent advances in surrogate modeling, such as reduced-order neural network approximations and reinforcement learning (RL)-based controllers, have shown promise in mitigating this issue by learning fast, near-optimal decision policies. RL architectures, in particular, can continuously improve dispatch and storage strategies through interaction with simulated or real environments, enabling adaptive and data-efficient optimization under uncertainty.

Another critical consideration lies in the interpretability and physical consistency of purely data-driven models. Conventional ML algorithms, such as long short-term memory (LSTM) networks, convolutional neural networks (CNN), are effective in capturing nonlinear relationships but often lack transparency, making them unsuitable for safety-critical or regulatory applications. This challenge has driven the rise of physics-informed and hybrid modeling approaches (Rangel-Martinez et al., 2021). Among these, physics-informed neural networks (PINNs) represent a significant advancement by embedding governing equations, such as mass balance, energy conservation, and electrochemical kinetics, directly into the network's loss function (Amiri et al., 2024). By constraining predictions with physical laws, PINNs ensure that outputs remain consistent with real-world behavior, even when data are scarce or noisy (Dong et al., 2024; Xia et al., 2024). Similarly, hybrid modeling frameworks that combine physics-based simulators (e.g., TRNSYS, HOMER, or MATLAB/Simulink) with ML modules enable bidirectional learning: physical models generate synthetic training data for ML, while ML enhances parameter estimation and uncertainty quantification within physics-based models.

In the context of HRES, these hybrid frameworks have proven particularly valuable for modeling dynamic behaviors of electrolyzers, fuel cells, and hydrogen storage systems under fluctuating renewable inputs. For example, a physical model-assisted AI can capture the transient response of proton exchange membrane electrolyzers or the degradation characteristics of fuel cells more accurately than purely empirical models (Ko et al., 2025). When coupled with multi-objective optimization algorithms such as NSGA-II or Bayesian optimization, this approach facilitates more precise scheduling and energy dispatch, balancing hydrogen production cost, system reliability, and environmental performance. Furthermore, integrating ML with uncertainty quantification methods (e.g., Bayesian neural networks or Gaussian processes) enhances the robustness of optimization under stochastic renewable generation and market volatility, ensuring that planning decisions remain reliable under varying policy and economic scenarios (Tan et al., 2022).

5. Conclusion

This review synthesizes recent advances in hydrogen-based hybrid renewable energy systems (HRESs), focusing on hydrogen energy storage integration, system modeling simulation tools and AI-driven optimization. A comparative analysis of modeling platforms (e.g., HOMER, iHOGA, TRNSYS, MATLAB/Simulink) and optimization algorithms highlights the superiority of intelligent, hybrid approaches for achieving multi-objective performance. Despite these advances, existing frameworks often overlook social–environmental factors and face challenges with data availability, model complexity, and interpretability. Incorporating physical principles into AI modeling can significantly enhance prediction reliability and transparency. Future research should focus on developing hybrid optimization models that combine stochastic, data-driven, and physics-based approaches to enable adaptive and explainable energy management in hydrogen-integrated renewable systems.

CRediT authorship contribution statement

Chenglong Li: Writing – original draft, Methodology, Conceptualization. **Tianqi Yang:** Writing – review & editing, Methodology, Funding acquisition. **Wenchao Cai:** Software, Formal analysis. **Kodjo Agbossou:** Writing – review & editing, Investigation, Conceptualization. **Pierre Bénard:** Visualization, Supervision, Resources. **Richard Chahine:** Supervision, Conceptualization. **Yi Zong:** Writing – review & editing, Supervision. **Yaze Li:** Visualization, Software, Resources. **Shenglin Su:** Visualization, Investigation, Data curation. **Guodong Li:** Visualization, Software, Investigation. **Xianglin Yan:** Resources, Investigation, Data curation. **Jin Li:** Investigation, Data curation. **Jinsheng Xiao:** Writing – review & editing, Validation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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